# Problematic Gaming Early Warning

## Data Preparation Feature Engineering and Model Exploration

Group 7 · SIC Capstone · Assignment 3 · 31 October 2026

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### Data Preparation and Feature Engineering

### 1 Overview

We investigate whether early gameplay records improve prediction of later Gaming Disorder Test (GDT) scores beyond the baseline GDT and recorded playtime. This submission implements cohort construction, an exploratory Steam feature table, and a completed nested cross-validation experiment. It continues the version and primary target specified in our concept note. The result is preliminary: detailed daily features do not improve mean absolute error over baseline GDT plus recorded hours in this experiment. Full platform coverage verification and secondary-target modeling remain subsequent stages.

### 2 Data Collection

We downloaded the cleaned survey_biweekly.csv.gz, survey_intake.csv.gz and steam.csv.gz files from Open Play, using commit eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840. This is the snapshot used in Assignment 2; newer releases are not mixed into this experiment. The compressed CSVs were read directly with pandas. The source and cleaning documentation describe pseudonymized surveys and gameplay records (Digital Wellbeing, n.d.).

The survey file contains 6,844 records from 1,948 people. Intake contains 33,046 screening records. Eligibility is qualified=True and age 18–40. We join by pid, retain scheduled waves 1–6, and require all four GDT items at both waves 1 and 6. This reproduces 710 people. For the first telemetry experiment, we retain 543 of these people with a recorded Steam hour wholly inside the early window. The remaining 167 are excluded from this experiment, not labeled as non-players.

### 3 Data Cleaning

Provider cleaning and our cleaning are distinct. We use the provider’s cleaned files rather than claiming to have reconstructed their raw-data pipeline. Our steps map Never/Rarely/Sometimes/Often/Very often to 1–5, require four nonmissing items before summing to 4–20, check person-wave uniqueness, and join intake eligibility. We do not impute GDT targets or treat off-schedule waves as extra measurements.

Timestamps are parsed in UTC. We retain only complete hourly buckets beginning at or after wave 1 and ending no later than wave 1 plus 14 days. This avoids using a final partial bucket extending beyond the prediction time. Provider-adjusted minutes are aggregated across titles. No additional winsorization is applied. High daily values are retained for this exploratory run; idle time, concurrent titles and hourly totals above 60 minutes require a separate coverage and plausibility audit before operational use.

# 4 Exploratory Data Analysis

| Selection step | People |
|---|---|
| Both complete GDT measurements before eligibility | 730 |
| Matched to intake | 723 |
| Qualified and aged 18–40 | 710 |
| At least one complete early Steam bucket | 543 |

Among the 710 eligible completers, 199 show an increase, 136 no change and 375 a decrease in GDT total. Mean GDT is 8.44 at baseline and 7.63 at wave 6. The average decline is descriptive and cannot be interpreted as an intervention effect. This matched cohort may be affected by attrition and selection.

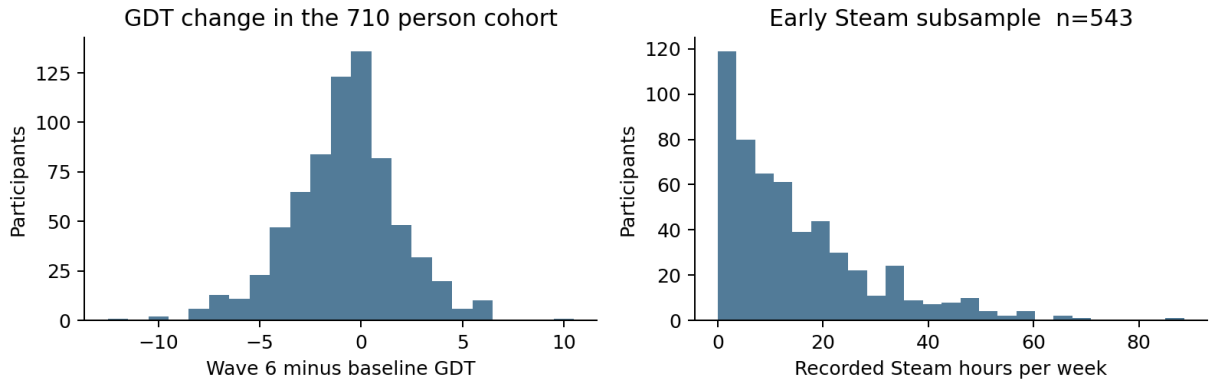



Figure 1. GDT changes in the survey cohort and recorded early Steam hours in the modeling subsample. Computed from the pinned data for this submission.

| Steam feature in the 543 person subsample | Median | 25th to 75th percentile |
|---|---|---|
| Recorded hours per week | 11.49 | 4.38–21.23 |
| Days with recorded play out of 14 | 9 | 5–13 |
| Daily recorded-minute standard deviation | 95.65 | 54.97–148.82 |
| Maximum recorded minutes in a study day | 316.37 | 172.61–512.64 |

The recorded-hours distribution is right-skewed: the mean is 15.04 hours per week, above the median. Baseline GDT averages 8.62 in this Steam subsample, compared with 8.44 in the 710-person cohort. The samples are therefore not interchangeable. We compare every model only on the same 543 people.

A day without a positive Steam record is represented as zero recorded minutes for aggregation. It is not verified zero gameplay: private profiles, untracked games or other platforms may be unobserved. Consequently, these features describe captured Steam activity and not complete gaming behavior. An 80% observation-coverage rule has not been applied, because the cleaned positive-play records alone do not establish hourly observability.

# 5 Feature Engineering

The prediction time is 14 days after the baseline survey. Each row represents one person. Fourteen study-day bins are defined by elapsed 24-hour periods from baseline, not local calendar days. This first experiment deliberately uses features that do not require local-time or exact-session reconstruction.

| Feature | Definition | Reason |
|---|---|---|
| baseline | Sum of four wave 1 GDT items | Strong starting symptom predictor |
| weekly_hours | Sum of captured minutes over 14 days divided by 120 | Comparable weekly amount of recorded play |
| active_days | Number of study-day bins with recorded minutes > 0 | Frequency of captured activity |
| sd_minutes | Sample standard deviation of 14 daily totals | Variation in recorded daily play |
| max_minutes | Largest daily recorded-minute total | Concentration of play in a single study day |
| week_change | Second 7-day total minus first 7-day total, in hours | Direction and size of early change |

The target is the wave 6 GDT total. GDT change is obtained by subtracting each person's baseline from the observed and predicted wave 6 totals. The absolute residual is identical for these two representations; $R^2$ is not, so the reported $R^2$ refers to wave 6 total. GDT's three-month recall periods overlap across measurements. We predict later measured symptom burden, not new disorder onset.

Night-time share, exact long-session measures, local weekday/weekend differences, game genres, Xbox/Nintendo features and NLP are not included in the completed experiment. These remain extensions of Assignment 2, requiring timezone, session and coverage validation. The secondary target remains next-survey routine interference, based on gaming_value_routines; no secondary-target performance is claimed here.

## 6 Data Transformation

Each learned estimator is inside a scikit-learn Pipeline: median imputation, StandardScaler and regression. Imputation and scaling are fitted only on the training portion of each inner or outer split. Although current engineered features are complete, the imputer provides a consistent pipeline. Standardization supports Ridge regularization; it is unnecessary for tree splits but is retained in the shared experimental pipeline. We do not fit transformations to the full sample before cross-validation.

Identifiers, timestamps, wave 6 survey answers and post-cutoff gameplay are excluded from model inputs. The GDT response order is explicitly encoded, rather than using arbitrary alphabetical category codes. No synthetic labels, oversampling, log transformation or clinical risk threshold is used.

# Model Exploration

## 1 Model Selection

| Model | Inputs | Purpose |
|---|---|---|
| M0 | Baseline GDT copied as prediction | No-change control |
| M1 Ridge | Baseline GDT | Learned symptom-only baseline |
| M2 Ridge | M1 plus weekly_hours | Added value of recorded amount |
| M3 Ridge | All six engineered inputs | Added value of daily patterns |
| M3 Gradient Boosting | Same six inputs | Small nonlinear comparison |

Ridge is suitable for a modest participant sample and correlated duration features. Its limitation is a linear response surface. Gradient boosting can capture nonlinear effects but can overfit this sample, so depth and leaf size are restricted. Elastic Net and Random Forest from the concept note remain candidates for the next iteration; they were not trained in this submission. Deep sequence models are outside the first-stage scope.

## 2 Model Training

We completed five outer folds using KFold(shuffle=True, random_state=42). Since the GDT table has exactly one row per person, row folds are also person folds. In each outer training set, three shuffled inner folds (random_state=43) choose hyperparameters by negative MAE. All models use identical outer splits. This is an exploratory estimate of generalization to unseen people, not a chronological deployment test. For the repeated-wave secondary target, GroupKFold by pid will be required.

Ridge alpha is selected from 0.1, 1 and 10. M1 and M2 select 0.1 in every outer fold; M3 selects 0.1 in four folds and 10 in one. GradientBoostingRegressor uses 100 trees, learning_rate=0.05, min_samples_leaf=20, random_state=42 and tuned max_depth in {1,2}. Selected depths are 1,1,2,2,1. Other parameters use the installed scikit-learn defaults. Hyperparameters are never selected using outer test scores.

Every reported prediction is produced for an outer-fold participant absent from that fold’s training set. The five held-out prediction blocks are concatenated to calculate the reported metrics. M0 has no fitted parameters. Feature definitions were fixed before this comparison; no feature search based on outer-fold scores was performed.

## Experimental scope

This run tests captured Steam activity conditional on at least one early recorded bucket. It is not the completed multi-platform design, and no minimum valid-observation coverage has been established. This selection may miss people who stopped playing or whose profiles were private. The next audit must distinguish those cases before a final early-warning claim is evaluated.

# 3 Model Evaluation

| Model | MAE | RMSE | R squared |
|---|---|---|---|
| M0 | 2.1031 | 2.8304 | 0.1920 |
| M1 | 1.8846 | 2.4057 | 0.4162 |
| M2 | 1.8757 | 2.3852 | 0.4262 |
| M3 | 1.8778 | 2.3893 | 0.4242 |
| M3 GB | 1.9168 | 2.4171 | 0.4107 |

Table 1. Pooled out-of-fold metrics on the same 543 people. Errors are in GDT total-score points; lower MAE/RMSE and higher $R^2$ are better.

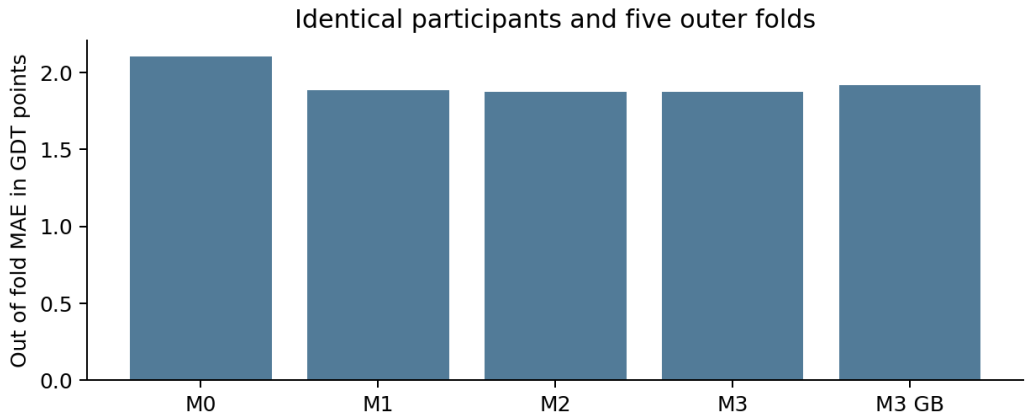



Figure 2. MAE for all completed models. This regression task does not produce a confusion matrix or ROC curve without defining a separate classification target.

M1 improves over copying baseline GDT: MAE falls from 2.1031 to 1.8846. Adding recorded hours gives M2 an MAE of 1.8757, a small descriptive reduction of 0.0089 points. Adding daily pattern features increases MAE slightly to 1.8778. Gradient boosting is also weaker than M2 in this run. These results do not establish useful extra predictive value from the tested daily features.

The participant-level paired difference MAE(M3) minus MAE(M2) is 0.0021 points. A percentile bootstrap of the fixed out-of-fold absolute-error differences, with 2,000 participant resamples and seed 42, gives an exploratory 95% interval [-0.0159, 0.0201]. It includes zero. This conditional interval does not repeat model fitting or capture all model-selection uncertainty, so it is not a definitive significance test.

No independent external test set, clinical calibration, SHAP analysis or deployment evaluation has been completed. We do not choose the best model and report its cross-validation score as an unbiased final deployment estimate. The present evidence supports continuing the coverage audit and testing well-defined timing features, not issuing individual risk alerts.

### Interpretation and next steps

Next, validate observation coverage, duration plausibility and local time before adding supported platform features. Repeat M2–M3 on identical eligible people and evaluate routine interference using person-grouped splits. NLP and dashboard work follow the core validation. These observational scale predictions establish neither causality nor diagnosis.

# 4 Code Implementation

The following excerpts show the implemented transformations and evaluation logic. Files are the cleaned CSVs under the pinned Open Play commit. The complete experiment was run for this submission; the numerical results above are measured outputs.

### Cohort and target construction

```
import pandas as pd
import numpy as np

b = pd.read_csv("survey_biweekly.csv.gz", low_memory=False)
i = pd.read_csv("survey_intake.csv.gz", low_memory=False)
labels = ["Never", "Rarely", "Sometimes", "Often", "Very often"]
encoding = dict(zip(labels, range(1, 6)))
g = b[[f"gdt_{k}" for k in range(1, 5)]]
b["gdt"] = g.apply(lambda c: c.map(encoding)).sum(
    axis=1, min_count=4)
q = b.merge(i[["pid", "qualified", "age"]], on="pid",
            validate="many_to_one")
q = q[q.qualified.eq(True) & q.age.between(18, 40)
      & q.wave.between(1, 6)]
p = q.pivot(index="pid", columns="wave", values="gdt")
p = p[[1, 6]].dropna()
p.columns = ["baseline", "target"]
p = p.join(q[q.wave == 1].set_index("pid")[["date"]])
p["date"] = pd.to_datetime(p.date, utc=True)
```

### Early window and recorded activity features

```
t = pd.read_csv("steam.csv.gz", usecols=[
    "pid", "datetime_hour_start", "minutes"])
t = t.merge(p[["date"]], on="pid")
t["time"] = pd.to_datetime(t.datetime_hour_start, utc=True)
t["day"] = np.floor((t.time - t.date).dt.total_seconds()
                    / 86400).astype(int)
cutoff = t.date + pd.Timedelta(days=14)
t = t[t.day.between(0, 13)
      & ((t.time + pd.Timedelta(hours=1)) <= cutoff)]
daily = t.groupby(["pid", "day"]).minutes.sum().unstack()
daily = daily.reindex(columns=range(14)).fillna(0)
x = p.join(daily, how="inner")
a = x[list(range(14))]
# Zero means no recorded play, not verified non-play.
x["weekly_hours"] = a.sum(axis=1) / 120
x["active_days"] = (a > 0).sum(axis=1)
x["sd_minutes"] = a.std(axis=1)
x["max_minutes"] = a.max(axis=1)
x["week_change"] = (a.iloc[:, 7:].sum(axis=1)
                    - a.iloc[:, :7].sum(axis=1)) / 60
```

# Code Implementation continued

## Nested validation for each model input set

```
from sklearn.pipeline import make_pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge
from sklearn.model_selection import KFold, GridSearchCV
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error, r2_score

features = ["baseline", "weekly_hours", "active_days",
            "sd_minutes", "max_minutes", "week_change"]
y = x.target.to_numpy()
outer = KFold(5, shuffle=True, random_state=42)
# Run this block for M1, M2, and M3 with the same folds.
input_sets = {"M1": features[:1], "M2": features[:2],
              "M3": features}
for name, columns in input_sets.items():
    predictions = np.empty(len(x))
    for train, test in outer.split(x):
        pipe = make_pipeline(SimpleImputer(strategy="median"),
                             StandardScaler(), Ridge())
        search = GridSearchCV(
            pipe, {"ridge__alpha": [0.1, 1, 10]},
            cv=KFold(3, shuffle=True, random_state=43),
            scoring="neg_mean_absolute_error", n_jobs=1)
        search.fit(x.iloc[train][columns], y[train])
        predictions[test] = search.predict(x.iloc[test][columns])
    print(name, mean_absolute_error(y, predictions),
          np.sqrt(mean_squared_error(y, predictions)),
          r2_score(y, predictions))
```

M0 directly uses x.baseline. For the gradient-boosting comparison, replace Ridge with GradientBoostingRegressor(n_estimators=100, learning_rate=0.05, min_samples_leaf=20, random_state=42), use all six inputs, and tune gradientboostingregressor__max_depth over [1,2]. The outer and inner split definitions remain unchanged.

## Paired uncertainty estimate

```
# pred_m2 and pred_m3 are aligned outer-fold predictions.
diff = np.abs(pred_m3 - y) - np.abs(pred_m2 - y)
rng = np.random.default_rng(42)
boot = [rng.choice(diff, len(diff), replace=True).mean()
        for _ in range(2000)]
print(diff.mean(), np.quantile(boot, [0.025, 0.975]))
```

Participant identifiers are used only for joins and split integrity. They are excluded from the exported modeling inputs and from figures. No participant-level timelines or recognizable text responses are published.

## References and reproducibility

The experiment retains the snapshot specified in the submitted Group 7 concept note. All reported counts, figures and scores were recomputed for this assignment. The choice of a Steam-only preliminary experiment is a bounded implementation step within the broader multi-platform plan, not a replacement of its research question.

Group 7. (2026). Capstone concept note and implementation plan: Problematic Gaming Early Warning [Course submission].

Digital Wellbeing. (n.d.). Open Play [Dataset and analysis code; commit eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840]. https://github.com/digital-wellbeing/open-play/tree/eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840

Ballou, N., Földes, T. A., Vuorre, M., Hakman, T., Magnusson, K., & Przybylski, A. K. (n.d.). Open Play: A longitudinal dataset of multi-platform video game digital trace data and psychological measures. https://digital-wellbeing.github.io/open-play/

Pontes, H. M., Schivinski, B., Sindermann, C., Li, M., Becker, B., Zhou, M., & Montag, C. (2021). Measurement and conceptualization of gaming disorder according to the World Health Organization framework: The development of the Gaming Disorder Test. International Journal of Mental Health and Addiction, 19, 508–528. https://doi.org/10.1007/s11469-019-00088-z

scikit-learn. (n.d.). Cross-validation: Evaluating estimator performance. https://scikit-learn.org/stable/modules/cross_validation.html

scikit-learn. (n.d.). Ridge. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Ridge.html

scikit-learn. (n.d.). GradientBoostingRegressor. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingRegressor.html

The data are released under CC0 with an additional no-reidentification clause. We retain only necessary research variables, do not link people to external identities, and present aggregate results. Access date for online documentation: 31 October 2026.

Execution environment: Python 3.12.14; pandas 2.2.3; NumPy 2.3.5; scikit-learn 1.8.0. CPU execution. Random seeds: outer folds 42, inner folds 43, gradient boosting 42, bootstrap 42.